# Miracle Agbebaku

Backend engineer: APIs, realtime and edge services

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## SUMMARY

Backend engineer who builds APIs and realtime services in Python and TypeScript: a FastAPI orchestration pipeline, a multi-account WhatsApp service and a Cloudflare edge backend with WebSockets and Durable Objects. Focus on validation at boundaries, isolation between accounts, rate limits and tests.

## PROJECTS

### ELLA

*Active development*

*TypeScript, Node.js, Express, Supabase, React, Vite*

Multi-account WhatsApp service built around NEXUS CORE.

- Runs many linked WhatsApp accounts in one Node.js and TypeScript process, with each account's session isolated and every chat, document and memory lookup scoped by account and chat.
- Persists account, session and preference data in Supabase with migrations, and built a provider gateway with ordered fallback and health tracking.
- Treats links and fetched pages as hostile input, blocking credentials, unusual ports, private addresses and unsafe redirects, with about 50 test scripts.

### Lagoon

*In development*

*JavaScript, Cloudflare Workers, Durable Objects, WebSockets, Playwright*

Cloudflare edge backend for a social platform: Workers, D1 and Durable Objects.

- Built live messaging over WebSockets with a per-user Durable Object that holds connections and delivers events, plus delivery acknowledgements, sync and history.
- Added chunked media upload with limits of 384 KB per chunk, 25 MB per file and 200 MB per person per day, and request protection with local and shared rate-limit counters.
- Enforced privacy and age rules on the server instead of the interface, with reports, suspensions and appeals leaving audit records.

### NEXUS CORE

*Working prototype*

*Next.js, React, TypeScript, FastAPI, Pydantic, SQLite*

FastAPI service for a governed AI agent pipeline.

- Built the API for a pipeline that validates model output with Pydantic and streams every stage to clients over Server-Sent Events.
- Stored projects, conversations, tasks and hash-linked audit records in SQLite, with full-text message search where the runtime supports it.
- Found and fixed a timestamp ordering bug: mixed ISO formats sorted wrongly as text, so every timestamp is now written with explicit microsecond precision.

## SKILLS

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| **Languages** | Python, TypeScript, JavaScript |
| **Backend and APIs** | FastAPI, Pydantic, Node.js, Express, Zod, Webhooks |
| **Realtime and edge** | Cloudflare Workers, Durable Objects, WebSockets, Server-sent events |
| **Data and auth** | PostgreSQL, SQLite, Supabase |
| **Testing and delivery** | Playwright, Vitest, GitHub Actions, Git, Render, Vercel |
| **Integrations** | WhatsApp, FFmpeg |
| **Practices** | input validation, account isolation, rate limiting, audit logging, automated testing |

## WRITING

20 engineering notes on trust boundaries, AI agent pipelines, messaging systems and how the portfolio is built, at miracle-agbebaku-portfolio.vercel.app/engineering.